

Experiment No 04

Aim:

To develop a blockchain-powered web application using Solidity on Remix IDE, connect it with MetaMask wallet, and deploy a smart contract on the Sepolia Testnet to record and retrieve student marks.

Theory:

Blockchain Technology

A blockchain is a distributed, immutable ledger maintained across a network of nodes rather than a single central server. Each block contains a set of transactions, a timestamp, and a cryptographic hash of the previous block, forming a tamper-evident chain. Once data is written to a block and confirmed by the network's consensus mechanism, it cannot be altered or deleted, which is what makes blockchains trustworthy for recording transactions without relying on a central authority.

Smart Contracts

A smart contract is self-executing program code deployed directly onto a blockchain. Once deployed, its logic runs exactly as written, and its execution cannot be changed, censored, or reversed by any single party — not even the contract's creator, unless the code explicitly allows it (e.g., via an owner-only function). Smart contracts have their own address on the blockchain, can store data permanently in "state variables," and expose functions that anyone can call to read or modify that data. Every function call that changes data is itself a transaction, requiring computational cost ("gas") and a cryptographic signature from the caller's wallet.

Solidity

Solidity is a statically-typed, contract-oriented programming language designed specifically for writing smart contracts on Ethereum-compatible blockchains. It supports familiar programming constructs — structs, mappings, modifiers, events, and access control — while enforcing rules unique to blockchain environments, such as explicit visibility for functions (public, private) and built-in overflow/underflow protection (from Solidity 0.8 onward). Contracts written in Solidity are compiled into EVM (Ethereum Virtual Machine) bytecode before deployment.

Remix IDE

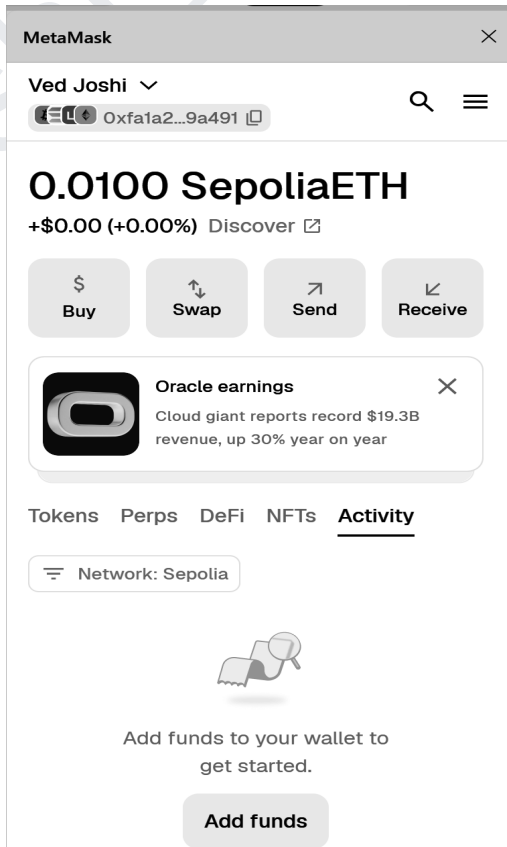
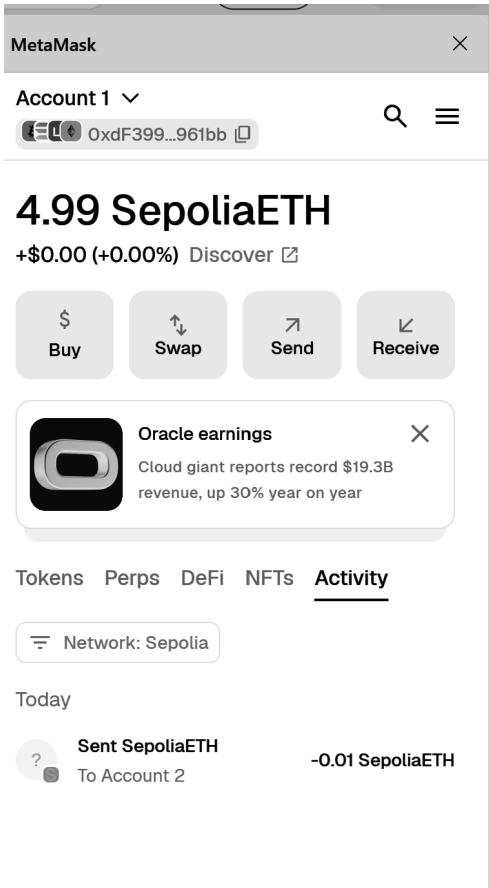
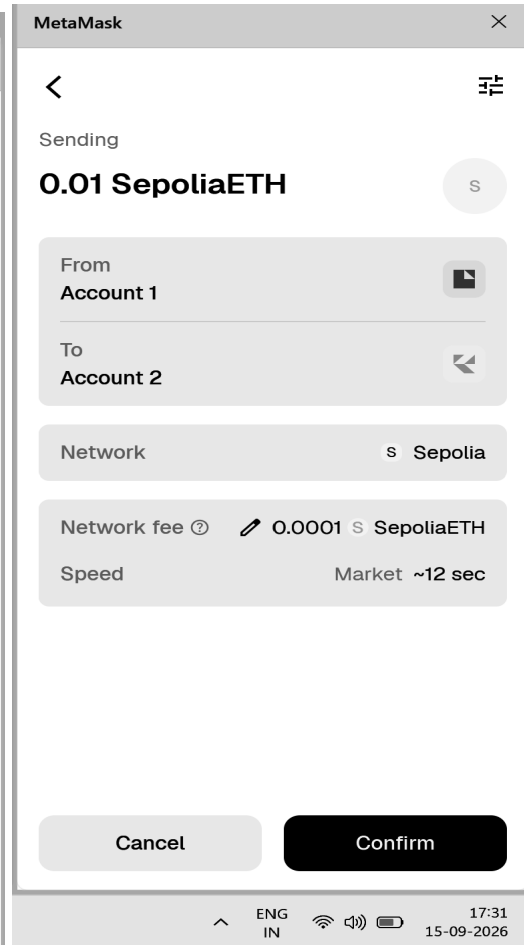
Remix is a free, browser-based Integrated Development Environment for writing, compiling, debugging, testing, and deploying Solidity smart contracts, with no local installation required. It provides a file explorer, an in-browser Solidity compiler, and a "Deploy & Run Transactions" module that can connect to different environments — a fully simulated in-browser blockchain (Remix VM), or a real network via an injected wallet provider like MetaMask.

MetaMask Wallet

MetaMask is a cryptocurrency wallet available as a browser extension that manages a user's private keys locally and never exposes them to the websites it interacts with. It acts as an intermediary between a dApp (like Remix) and the blockchain: whenever a transaction needs to be sent — whether deploying a contract or calling a function that modifies data — MetaMask intercepts the request, displays its details (network, gas fee, destination address), and requires explicit user confirmation before signing and broadcasting it. This "injected provider" model is what lets browser-based tools like Remix send real transactions without ever handling the user's private key directly.

Testnets and Gas

A testnet (such as Sepolia) is a separate blockchain network that mirrors the behavior, rules, and tooling of the Ethereum mainnet but uses worthless test tokens instead of real cryptocurrency. Developers obtain free test ETH from "faucets" to pay for gas — the computational fee required to execute transactions and store data on-chain — allowing contracts to be built, deployed, and tested realistically without financial risk. Once a contract is confirmed to work correctly on a testnet, the same code can be deployed to the mainnet with real funds.



Sepolia Testnet

Search by Address / Txn Hash / Block / Token

Transaction Details

Sponsored: Etherscan API - Onchain data for your agents across 60+ chains via MCP, CLI, Skills. [Build with AI.](#)

Overview State

TRANSACTION ACTION
0xdF399bc7...a9D1961bb Transfer 0.01 ETH to 0xfa1a2E7d...C6049a491

[This is a Sepolia Testnet transaction only]

Transaction Hash: 0x1fe68869d06d1ec0645eff98e5841550948d09063603a98d14fdc089a78c1b6

Status: Success

Block: 11709744 6 Block Confirmations

Timestamp: 1 min ago Sep-15-2026 12:02:00 PM +UTC

From: 0xdF399bc7ceEd56fA757A2351078f07a9D1961bb

To: 0xfa1a2E7dAAAa16dA02E4Df4A8CeA031C6049a491

Value: 0.01 ETH

Transaction Fee: 0.000056048835423 ETH

Gas Price: 2.668992163 Gwei (0.000000002668992163 ETH)

REMIX 2.5.7 Learneth Solidity Beginner Course

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DEPLOY & RUN TRANSACTIONS

Environment: Sepolia Testnet - MetaMask

Account 16 4.989 ETH

Deploy Sepolia (11155111)

TRUSTescrow TRUSTescrow.sol

influencer address 0xfa1a2E7dAAAa

Call data Parameters

```
1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.3;
3
4 contract TRUSTescrow {
5
6     // Address of the advertiser who creates and funds the campaign
7     address payable public advertiser;
8
9     // Address of the influencer who receives the campaign payment
10    address payable public influencer;
11
12    // Amount of ETH currently locked in escrow
13    uint public escrowAmount;
14
15    // Indicates whether the campaign has been completed
16    bool public campaignCompleted;
17
18    // Constructor
19    // The account deploying the contract becomes the advertiser
20    // The influencer's wallet address is provided during deployment
21    constructor(address payable _influencer) {
22        infinite gas 533600 gas
```

Account 1

4.99 SepoliaETH

+\$0.00 (+0.00%) Discover

Buy Swap Send Receive

Oracle earnings

Cloud giant reports record \$19.3B revenue, up 30% year on year

Tokens Perps DeFi NFTs Activity

Network: Sepolia

Today

Sent SepoliaETH -0.01 SepoliaETH

Remix is moving to app.remix.live. Your Workspaces are stored the browser and won't be accessible from the new URL unless you follow the move instructions.

Move my Workspaces

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DEPLOY & RUN TRANSACTIONS

Environment: Sepolia Testnet - MetaMask

Account 16 4.989 ETH

Verify Contract on Explorers

influencer address 0xfa1a2E7dAAAa

Call data Parameters

Value 0 wei

Gas limit auto 0

Deploy

Deployed Contracts

Interact with a deployed contract

```
1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.3;
3
4 contract TRUSTescrow {
5
6     // Address of the advertiser who creates and funds the campaign
7     address payable public advertiser;
8
9     // Address of the influencer who receives the campaign payment
10    address payable public influencer;
11
12    // Amount of ETH currently locked in escrow
13    uint public escrowAmount;
14
15    // Constructor
16    // The account deploying the contract becomes the advertiser
17    // The influencer's wallet address is provided during deployment
18    constructor(address payable _influencer) {
19        infinite gas 533600 gas
```

0 Listen on all transactions Filter with transaction hash or address

Welcome to Remix 2.5.7

Your files are stored in IndexedDB, 644.37 KB / 10 GB used

You can use this terminal to:

- Check transactions details and start debugging.
- Execute JavaScript scripts:
 - Input a script directly in the command line interface
 - Select a Javascript file in the file explorer and then run 'remix.execute()' or 'remix.executeCurrent()' in the command line interface
 - Right-click on a Javascript file in the file explorer and then click 'Run'

MetaMask

Account 1

Deploy a contract

This site wants you to deploy a contract

Estimated changes No changes

Network Sepolia

Request from remix.ethereum.org

Network fee 0.0017 SepoliaETH

Speed Market ~12 sec

Cancel Confirm

Explain contract

AI copilot

0

☐ Listen on all transactions

Filter with transaction hash or address

value: 0 wei data: 0x608...9a491 logs: 0 hash: 0xc10...04cf3

Debug

[Verification] Contract deployed. Checking explorers for registration...

Etherscan verification skipped: API key not provided.

Please input the API key in Remix Settings - Connected Services OR Contract Verification Plugin Settings.

[Blockscout] Verification submitted. Awaiting confirmation...

[Routescan] Verification submitted. Awaiting confirmation...

[Sourcify] Verification Successful! View Code

>

Did you know? You can use the help of AI for Solidity error, click on 'Ask RemixAI'.

RemixAI Copilot (enabled)

Deployed Contracts 1 + Add Contract

Interact with a deployed contract

TRUSTEscrow Sepolia 31s ago

0x3b7...0B04f

Functions

Select a function to interact with...

Low level interaction

Balance 0 ETH

MetaMask

Account 1

Transaction request

Balance changes

You send - 1 SepoliaETH

Added protection Optional

Because you're interacting with an unknown address, protection can prevent some malicious transactions. Learn more

Network Sepolia

Request from remix.ethereum.org

Interacting with 0x3b71C...0B04f

Network fee 0.0005 SepoliaETH

Speed Market ~12 sec

Cancel

Confirm

